

Philip Grayson

Principal Engineer / Tech Lead - DevOps, Java, AWS & Kafka

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Java and DevOps specialist who owns software end to end: designing, developing, testing, deploying, instrumenting, and debugging. A decade building and running high-throughput event pipelines, software systems, and AWS infrastructure.

Work Experience

Tech Lead / Principal Engineer

PokerStars (Flutter Entertainment)

Apr 2024

Built and maintained ETL systems using Kafka and Kafka Streams in Java. Focus on performance, multi-tenancy, and observability. Ran a team of ~3 developers and ~2 QA developers.

- Owned the team's technical decisions and software development. Worked with Product Owner and Architecture to turn requirements into solutions. Maintained team's work pipeline (planning, elaboration, estimation, unblocking).
- Hands-on across design, development, and test: implemented critical functionality, unit/integration tests, and BDD (gherkin). Designed deployment pipelines to AWS EKS, with strong focus on IAM (cross-account, principle of least privilege).
- Designed and implemented a 5x increase in Kafka record throughput through batching, while maintaining at-least-once delivery semantics; including developing load testing software and ~RFC documentation.
- Integrated with Prometheus/Grafana/AlertManager and stood up a self-service Kafka-UI instance for staff to inspect topic data directly.
- Performed Kafka capacity planning (using AWS MSK Open Monitoring), identifying CPU throttling (burstable T instances), and a needed 10x increase in storage capacity to handle a new feature.

Principal Engineer / DevOps

PokerStars (Flutter Entertainment)

Nov 2021 - Apr 2024

Deploying, maintaining, scaling software systems to process sporting event updates, to drive bet pricing and settlement. Peak 2,500+ updates per second.

- Focus on event processing latency, since any delays could allow customers to bet on known outcomes. Visualisation and monitoring of latency SLIs (using percentiles), and built break-glass emergency processes in case of unacceptable latency.
- Established myself as the go-to expert for Apache Storm and Apache Cassandra. Covering capacity planning, scaling, and version upgrades.
- Described software deployments and redeployment strategies (~immutable EC2 instances, stateful data migration, EBS volume management) using AWS SDK (CloudFormation).
- Led investigation and remediation of several incidents, often requiring close examination of application logs (from distributed systems), metrics and reading software source code. Non-production incidents were treated in the same way to help drive operational excellence. Root causes included Java GC pauses, DNS caching, Apache Storm internals, NTP backward time shift.
- Created and maintained comprehensive systems documentation, guides, and presentations, to help share knowledge with other technical members.
- Coaching on software systems and incident investigation, including members outside my immediate team.

Automation Engineer / Tech Lead / Principal Infrastructure Engineer

Sky Betting & Gaming

Sep 2015 - Nov 2021

Managing platform infrastructure like Confluence, Jira, Bitbucket, Jenkins, Kafka, Artifactory, Chef.

- Built and maintained several Apache Kafka clusters - including custom-written infrastructure-as-code (in Chef), user documentation, monitoring and automated alerting, version upgrades. 2+ billion records a day, 10+TB storage used.
- Planned and performed high risk stateful system migrations and upgrades (for Confluence, Jira, Bitbucket). Optimising migration speed (reduced a planned Bitbucket downtime from 2.5 hours to 5 minutes). Built staging environments to practice migrations using production data set.
- Progressed to tech lead and became responsible for the operational running and technical vision of services maintained by the team, advising on squad roadmap, handling incidents, building user relationships, and mentoring more junior team members in programming.
- Developed custom Kubernetes Operator for Percona XtraDB clusters. Authored impact assessment and HLD for a 40+TB Elasticsearch deployment.

Junior Developer / Developer

Ebuyer

Aug 2011 - Sep 2015

Frontend (HTML, XSL, jQuery) and backend (PHP, SQL) developer

- Migrated payments backend, replacing cardholder data with tokenisation for PCI DSS.
- Lead developer on various frontend (customer facing) navigation changes, including header, banners and faceted search.
- Build backoffice reports to visualise various data, with interactive graphs and data tables.
- Took 1 year off to complete my BSc Software Development degree.

Skills

Languages:

Java

Shell

Previously used Go / Ruby / Python

Frameworks & Data:

Spring Boot

Kafka (Server and Client, Self-hosted, AWS MSK, Confluent Cloud)

Databases (SQL, Cassandra, Elasticsearch)

Cloud & Infrastructure:

AWS (SysOps Associate)

Kubernetes (AWS EKS, Operators)

Helm/Kustomize

Terraform

Docker

Linux

GitHub Actions

Jenkins

Git (inc. advanced usage like rebase, cherrypick, reflog)

Reliability & Observability:

Prometheus

Grafana

Capacity planning

Incident investigation

Soft skills:

Communication: Written / oral / visual

Independent working

Fast learning of software systems / libraries / languages

Certificates

AWS Certified SysOps Administrator - Associate

2025-01

BSc (Hons) Software Development, University of Huddersfield

2013